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Sensing the World Reflection 3 (Project):

The project has been my favorite part of the class so far! For this section of the module, we were tasked with creating a system that would be a prototype for a solution to a problem in the real world. Our guidelines for this project was that human interaction was necessary, the system would provide a service or support to a situation so as to improve the quality of life for an individual, there needed to be multiple inputs and outputs for our system, and we needed to use AI to help develop our code. We also needed to include at least 1 LED, the homemade sensor, one additional input element, and at least one motor in our system. We were then given access to the key site lab space along with a set of materials which include Knex, cardboard, string, pipe cleaners, popsicle sticks, tape, hot glue, and straws. Our group brainstormed a possible system that we could create using the moisture sensor we had created in class, and we ultimately landed on making an automated umbrella system. This system would automatically go up when it detected moisture so that the user would be protected from the elements. This could help those who are either elderly or disabled and may have difficulty opening an umbrella or holding one for extended periods of time. The main part of the project which I worked on was the actual umbrella part and the more mechanical side of the system. I selected the materials to use which included Knex, duct tape, straws, and hot glue. Then I took time to figure out which Knex pieces I wanted to use and how to actually create the Umbrella shape with the pieces. Eventually I was able to use a tire rim Knex with some other normal Knex pieces to create the general shape of an umbrella. I then wrapped this in duct tape to actually create the umbrella part. I then spent time deciding which material to use for the central shaft. I decided to use a mid-sized Knex because the long ones were too flimsy and could not hold up the actual umbrella part while the small ones did not have enough space underneath the umbrella. We then glued this umbrella to a mechanical arm connected to a servo motor so that when the servo motor was given the signal the arm would rotate up 90 degrees and the umbrella would be able to protect the area underneath from getting wet. Then once the rain had stopped there was a push button switch which when pressed would rotate the umbrella back to its resting position. While I was working on the umbrella my groupmates were working on the hardware and software. One of my teammates worked with AI to write the code. He did this by providing the AI with instructions on how to write the code so that when a certain moisture level (we measured this by ADC) was reached the LED would start flashing and the servo motor would rotate the mechanical arm up to 90 degrees. While he was doing this my other teammate wired the different components on the breadboard so the code could run and function correctly meaning the servo motor and the push button would actually function. We tested this out and luckily the system worked,

the umbrella was not too heavy, and it successfully protected the area underneath. For the rest of the time, we then worked to develop our presentation to effectively communicate our ideas. We split up the work so that everyone created a portion of the presentation. We then divided up which slides people were presenting on, usually by what part of the project that person worked on and then got ready to present. I prepared by creating notes for what I wanted to talk about on my slides just to make sure when I was practicing, I was hitting all the key points I wanted to talk about. Then finally we wrapped up today by presenting which went well and was a super fun experience because I got to share about this project which I had so much fun creating.

As previously mentioned, I really enjoyed this part of the module, it definitely had its challenges, but it was overall just a great experience. One of the hardest things was figuring out the moisture sensor. If we did not have the wires perfect it would not read at all or read levels that were way above the expected level of ADC. Also, near the end we broke our original moisture sensor by spraying it with water too much and wiping off the coating, so we needed to use another sensor which was giving us different ADC outputs than before. The lesson we learned here is that when working with finicky devices such as our moisture sensor you need to handle them gently so that you do not disrupt their functionality. We also should have listened to the direction of not putting direct water on the sensor. I think a major skill which I learned and developed over this project was teamwork. In a project like this it is impossible to do it all yourself, which was a challenge for me because for much of my high school education I would do all or almost all of group projects by myself because I didn't trust my groupmates to do it how I wanted. Therefore, it took some time to learn how to trust my groupmates to do parts of the project. However, I realized that teamwork is better because we were able to complete our prototyping way faster, it allowed all of us to become better and more knowledge at one part of this project, and it is less stressful to have to only do part of a project compared to the whole thing. Overall, I really enjoyed this module because it tested my abilities in fields which I am not as knowledgeable in and forced me to learn more about them. It also gave us a great opportunity to think creatively and have fun!